# [P2] The eleven-card lemma

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## Abstract

Every pairwise intersecting 6-uniform family of eleven cards has a transversal of size at most four. The proof splits on the maximum symbol degree: degree five or more is closed by pairing, degree four by the finite $K_7$ lemma (which has a hand proof in [P1]), and degree at most three by reduction to a short statement about 3-element sets on eleven points, checked by a SAT solver in a few seconds; an earlier exhaustive search over 5,373 instances is kept as a second check. The fifteen-card exclusion then follows in two lines, and the same lemma pins the maximum degree at exactly four in the sixteen-card stage of [P3].

Computer-assisted; not independently reviewed. No priority claimed. See the note at the end.


## 1. Why a second route

[P1] reaches $g(6) \geq 16$ through a degree histogram and thirteen hand eliminations. The route works but it is long, and it does not generalise: at sixteen cards the histogram stage needs an inequality whose right-hand side is no longer controlled, because the disjointness count $D$ of high-symbol pairs has no usable upper bound there.

The eleven-card lemma avoids the histogram entirely. It is a statement about eleven cards with nothing assumed about degrees, and everything else drops out of it.

## 2. The lemma

**Theorem 1.** *Every pairwise intersecting 6-uniform family of eleven cards has a transversal of size at most four.*

Suppose not: eleven cards, pairwise intersecting, and $\tau > 4$.

### Branch A — some symbol has degree at least five

Choose it. At most six cards remain; pair them and take one shared symbol per pair, at most three symbols. Total at most four, contradiction. So the maximum degree is at most four.

### Branch B — some symbol has degree four

Let $y$ have degree four and delete its four cards, leaving seven cards $R$.

$\tau(R) > 3$, since otherwise $y$ together with a three-cover of $R$ is a four-cover.

Every symbol of $R$ has residual degree at most two: a symbol $z$ of residual degree at least three would leave at most four cards, coverable by two more symbols, so $z$ plus those two is a three-cover of $R$.

Each card of $R$ must meet the other six through its six symbols, and each symbol reaches at most one other card. So every symbol of $R$ has residual degree exactly two and every pair of cards of $R$

meets in exactly one symbol. The seven cards are therefore the vertices of a $K_7$ and their 21 shared symbols are its edges.

Each of the four deleted cards contains $y$, which lies outside $R$, and must meet all seven cards of $R$. Its remaining five symbols therefore induce an edge cover of that $K_7$ of size at most five.

**Lemma 2 (the $K_7$ lemma; [P1, §3] proves it by hand).** *Given any four edge covers of $K_7$, each of size at most five, there is a four-edge cover of $K_7$ meeting each of them in an edge. Repetitions among the inputs are allowed.*

The four symbols of that four-edge cover then cover all seven cards of $R$ (an edge cover covers every vertex) and also the four cards of $y$ (each is met in an edge). That is a four-cover of all eleven cards, contradiction.

The lemma is verified exhaustively: 840 inclusion-minimal inputs, 315 four-edge covers, one fixed representative per isomorphism type, 294,239,817 quadruples certified, zero counterexamples. See `code/verify_k7.py` and [P1, §3] for the enumeration of the three minimal types.

### Branch C — every symbol has degree at most three

Write $(t, b, s)$ for the numbers of symbols of residual degree three, two and one in a card. Then $t + b + s = 6$, and since a card must meet the other ten, $2t + b \geq 10$. The only solutions are

$$A = (4, 2, 0), \quad B = (5, 0, 1), \quad C = (5, 1, 0), \quad D = (6, 0, 0).$$

Their excess degrees — the sum over other cards of intersection size minus one — are $0, 0, 1, 2$. Let $a, b, c, d$ count them, so $a + b + c + d = 11$, $c$ is even, and $4a + 5b + 5c + 6d$ is divisible by three.

Let $E$ be the multigraph on card pairs whose multiplicity is the intersection size minus one, and $F$ the graph with one edge per degree-two symbol. Degrees of $E$ on types $A, B, C, D$ are $0, 0, 1, 2$; degrees of $F$ are $2, 0, 1, 0$. $F$ is simple: a repeated $F$ edge needs two endpoints of $F$-degree at least two, hence of type $A$, whose $E$-excess is zero.

The degree-three symbols must decompose the pair multigraph

$$M = K_{11} + E - F$$

into triangles, repeats permitted. And that decomposition can contain no matching of size three: three vertex-disjoint triangles cover nine cards with three symbols, and the remaining pair is covered by one more, giving a four-cover.

**Petals.** Fix a card $x$ with $t$ degree-three symbols and $b$ degree-two symbols. Its degree-two symbols reach $b \leq 6 - t$ other cards, so its triangles reach at least $10 - b \geq t + 4$ other cards. In the link of $x$ (one pair of cards for each of its triangles) there are $t$ edges on at least $t + 4$ vertices, hence at least four components, hence four pairwise disjoint edges. So $x$ lies in four triangles that pairwise meet only in $x$; call them petals of $x$. Also $x$ lies in at most six triangles.

**Lemma 3 (computer-assisted).** *Let $H$ be a set of 3-element subsets of an 11-element set in which every point is the centre of four petals, and let $x$ be a point in at most six members of $H$. Then for some petal $P$ of $x$ there are two disjoint members of $H$ disjoint from $P$.*

Applied to the triangles (repeated triangles counted once, which changes neither petals nor the bound six), Lemma 3 gives three vertex-disjoint triangles, which is impossible. This closes Branch C.

`code/verify_eleven_cards_petals.py` checks Lemma 3 with a SAT solver in a few seconds, with $x$ and its petals fixed as 0 and $\{0,1,2\}$, $\{0,3,4\}$, $\{0,5,6\}$, $\{0,7,8\}$. The bound six is used: with seven the same model has a solution. On ten points it has a solution too. The script checks both, so the lemma has no slack in either parameter. Lemma 3 is also proved in Lean 4 (`lean/Petals.lean`, theorem `petals_11_6`), where `bv_decide` checks the solver's certificate with a checker verified in Lean; see `lean/README.md`. A hand proof is not known to us.

**A second check.** The first version of this branch searched the triangle decompositions directly, as follows. It is kept as an independent check of Branch C.

Every $F$ component is a path with $C$ endpoints and $A$ interiors, or a cycle of $A$ vertices of length at least three. Every $E$ component is a path with $C$ endpoints and $D$ interiors, or a cycle of $D$ vertices of length at least two, a two-cycle being a double edge. Type $B$ is isolated in both.

`code/verify_eleven_cards.py` generates every relative arrangement of $F$ and $E$ by labelling the $C$ vertices first, fixing their pairing in $F$, distributing $A$ interiors in nondecreasing numbers along the paths, partitioning the remaining $A$ vertices into cycles of length at least three, and then enumerating every $E$ pairing of the $C$ vertices, every allocation of $D$ interiors and every partition of the remaining $D$ vertices into cycles of length at least two. Duplicates are harmless.

That gives 5,373 instances across 58 non-empty profiles. Each one is searched exactly: at each step the solver takes a pair with positive remaining multiplicity, branches over every triangle containing it with positive capacity, and forbids a triangle that would complete a matching of size three with the triangles already chosen. All 5,373 return UNSAT.

Branches A, B and C are exhaustive, so Theorem 1 holds. □

## 3. Fifteen cards, in two lines

**Corollary 1.** *No pairwise intersecting 6-uniform family of at most fifteen cards has transversal number six. Hence* $g(6) \geq 16$.

*Proof.* Suppose fifteen cards with $\tau = 6$. If every degree were at most three, a fixed card's six symbols would meet at most $6 \cdot 2 = 12$ other cards, while fourteen must be met. So some symbol $x$ has degree at least four. Delete its cards: at most eleven remain, padded up to eleven by repeating existing cards if necessary, which changes no transversal number. Theorem 1 gives a four-cover of them, and with $x$ that is a five-cover. Smaller families are handled by the same padding. □

Compare [P1], where the same statement costs a weighted inequality, a count of thirteen histograms, and nine pages of incidence arguments, but uses no computer search.

## 4. What the lemma gives the sixteen-card stage

**Corollary 2.** *In a pairwise intersecting 6-uniform family of sixteen cards with* $\tau = 6$*, the maximum symbol degree is exactly four.*

*Proof.* The same local count gives a symbol of degree at least four, since a card must meet fifteen others and six symbols of degree at most three reach only twelve. A symbol of degree at least five would leave at most eleven cards, which Theorem 1 covers with four symbols, giving a five-cover. □

That is the hypothesis [P3] starts from.

## 5. Scope

Theorem 1 is sharp at eleven. `code/verify_core.py` (working directory, not part of the release checks) exhibits a twelve-card pairwise intersecting 6-uniform family with $\tau = 5$, so the lemma does not extend to twelve by raising the count alone.

An eight-card analogue does not hold either. There is an explicit eight-card pairwise intersecting 6-uniform family with maximum degree three and $\tau = 4$: take card 7 as the centre of four triangles $\{7, 0, 1\}$, $\{7, 0, 2\}$, $\{7, 3, 4\}$, $\{7, 5, 6\}$, give card 7 two pendant symbols, and join every remaining pair of cards directly by a degree-two symbol. All 1,330 three-symbol subsets fail to cover, and the four triangle symbols cover. This is why [P3] must enumerate eight-card cores rather than exclude them, and it is the reason the sixteen-card stage needs a SAT layer at all.

## 6. Verification

| claim | script |
|---|---|
| Lemma 2, all 294,239,817 quadruples | `code/verify_k7.py` |
| Lemma 3, with its two controls (bound seven; ten points) | `code/verify_eleven_cards_petals.py` |
| Lemma 3 in Lean 4 | `lean/Petals.lean` |
| Lemma 2 in Lean 4 | `lean/K7.lean` |
| Branch C again, all 5,373 instances UNSAT | `code/verify_eleven_cards.py` |

Branches A and B are hand arguments on top of Lemma 2, which has a hand proof in [P1, §3]. In Branch C the reduction to Lemma 3 is a hand argument and Lemma 3 is a SAT check.

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*This work was prepared with AI assistance (ChatGPT, OpenAI; Claude, Anthropic), used for algebraic derivation, for drafting and rewriting code and text, for running and re-running the computations, and for auditing the papers against their own scripts. The same tools were used in the review, so the review carries the same caveat. All statements were checked by the author, who is responsible for them. No priority is claimed and the result has not been independently reviewed by a human or a proof assistant; the repository README sets out the division of labour and the open obligations in full.*

## References

1. J. Barát and I. M. Wanless, *Intersecting and 2-intersecting hypergraphs with maximal covering number: the Erdős–Lovász theme revisited*, J. Combin. Des. 29 (2021), 260–286; arXiv:2011.04444. Lemma 2.2 there is the standard "choose a high-degree symbol and pair the rest" step used in Branch A.